(i) Karl – Person coefficient of skewness

According to Karl-Pearson the absolute measure of skewness = Mean – Mode.

Karl- Pearson coefficient of skewness = $\frac{Mean - Mode}{S.D}$

Example 6.16

From the known data, mean = 7.35, mode = 8 and Variance = 1.69 then find the Karl- Pearson coefficient of skewness.

Solution:

$$\text{Karl- Pearson coefficient of skewness} = \frac{Mean - Mode}{S.D}$$

$$\text{Variance} = 1.69$$

$$\text{Standard deviation} = \sqrt{1.69} = 1.3$$

$$\text{Karl- Pearson coefficient of skewness} = \frac{7.35 - 8}{1.3}$$

$$= \frac{-0.65}{1.3} = -0.5$$

(ii) Bowley's coefficient of skewness

In Karl Pearson method of measuring skewness the whole of the series is needed. Prof. Bowley has suggested a formula based on position of quartiles. In symmetric distribution quartiles will be equidistance from the median. $Q_2 - Q_1 = Q_3 - Q_2$, but in skewed distributions it may not happen. Hence

Bowley's coefficient of skewness (SK) = $\frac{Q_3 + Q_1 - 2Q_2}{Q_3 - Q_1}$

Example 6.17

If $Q_1 = 40$, $Q_2 = 50$, $Q_3 = 60$, find Bowley's coefficient of skewness

Solution:

Bowley's coefficient of skewness

$$= \frac{Q_3 + Q_1 - 2Q_2}{Q_3 - Q_1}$$

Bowley's coefficient of skewness

$$= \frac{40 + 60 - 2 \times 50}{60 - 40} = \frac{0}{20} = 0$$

$\therefore$ Given distribution is symmetric.

NOTE

If the difference between the mean and median or mean and mode is greater, the data is said to be more dispersed.

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